MILITARY SPECIFICATION

NETWORK, IMPEDANCE MATCHING

QUADRATURE HYBRID CU-1796( )/ARC-114

1. SCOPE

1.1 This specification covers one type of hybrid network which operates within the frequency range of 30 to 76 Megahertz (MHz) and functions as an impedance matching and phasing device as part of a homing antenna system and is designated, Network, Impedance Matching Quadrature Hybrid CU-1796( )/ARC-114. (See 6.1 and 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Material Handling, Wood, (General Construction Requirements)
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, corrugated and Solid Sheet Stock (Container Grade)
PPP-P-291	Paper, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive, Filament Reinforced
QQ-S-781	Strapping, Flat Steel

MILITARY

MIL-P-116	Preservation, Methods of
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-43014	Box, Water Resistant Paper Board, Folding, Setup and Metal-Stayed

FSC 5915

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids Runners, or Pallet-Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWING

USAECOM

SC-C-729979	Network, Impedance Matching Quadrature Hybrid CU-1796( )/ARC-114

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer. The title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. - CU-1796( )/ARC-114 is a hybrid network which functions as an impedance matching and phasing device as part of a homing antenna system. The network receives radio frequency signals from Network, Impedance Matching CU-1794( )/ARC-114 and applies them through a coaxial cable to Radio Set AN/ARC-114 homing inputs.

3.2 Performance. - The network on order shall meet the requirements of this specification.

3.3 Size and Weight. - The network outline dimensions shall be in accordance with SC-C-729979 and the total weight shall not exceed 175 grams.

3.4 First Article. - The contractor shall furnish First Article units when specified in the contract or purchase order. (See 4.3 and 6.2)

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3.5 Cleaning -

3.5.1 Parts - Prior to the application of conformal coating, on those parts affected, and after fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Post cleaning assembly processes shall minimize handling by bare hands to prevent contamination by fungus nutrients. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, and other parts.

3.5.2 Units - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.6 Finish, protective (See 4.4). - The units shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as screws, etc. and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be Film E in accordance with MIL-F-14072.

3.7 Securing of parts. -

3.7.1 General. - Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7.2 Securing of parts by threaded fasteners. - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.8 Marking (See 4.4). -

3.8.1 General. - Marking shall conform to MIL-M-13231.

3.8.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the unit(s).

3.8.3 Serial numbers. - The following unit(s) shall be serial numbered:

Network, Impedance Matching Quadrature Hybrid	CU-1796( )/ARC-114

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3.9 Soldering. -

3.9.1 Soldering. - Soldering shall conform to Requirement 5 of MIL-STD-454.

3.9.2 Rosin-base residues. - Flux residues shall be removed by applying organic solvents. Mechanical means such as agitation, brushing, etc. may be used in conjunction with the solvents. The cleaning solvents and methods used shall not have any deleterious effect on the parts, materials, and assemblies being cleaned.

3.9.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an avoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the unit(s); flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring. - Wiring shall be neat and sturdy.

3.10.1 Clearance. - Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.2 Splicing. - Wires in a continuous run between two (2) terminals shall not be spliced during the wiring operation.

3.10.3 Connections, general. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.11 Tropicalization of materials. - Materials shall be treated as follows:

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3.11.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.11.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.11.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.11.4 Statement of treatment. - The contractor shall submit for approval, to the Contracting Officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use.

3.12 Service conditions (See 4.7). - The network shall meet the service conditions for Class 1 equipment as specified in MIL-STD-810, unless otherwise specified.

3.12.1 Temperature-Altitude (See 4.7.1). - The network shall meet the electrical requirements of 3.13.1 through 3.13.5.2 during and after subjection to temperature-altitude test under the following conditions:

a. Non-operating. - Non-operating minimum exposure of 24-hours at -65°F at atmospheric pressure and minimum of 4-hours exposure at +160°F at atmospheric pressure.

b. Operating. - The network shall meet the full requirements of 3.13.1 through 3.13.5.2 in the range of -25°F through +131°F. Operable in the range of -65°F through -25°F, and the parameters in 3.13.1 through 3.13.5.2 shall be: Isolation input not less than 10 dB, isolation output not less than 9 dB, VSWR not to exceed 3:1, insertion loss no greater than 2.2 dB, amplitude levels no greater than 1.50 dB, input phase difference 90 $\pm$ 14°, output phase difference 180 $\pm$ 10° and 0 $\pm$ 10°.

3.12.2 Humidity (See 4.7.2). - The network shall meet the electrical requirements of 3.13.1 through 3.13.5.2 during and after subjection to humidity test of Procedure II, Method 507 of MIL-STD-810.

3.12.3 Vibration (See 4.7.3). - The network shall meet the electrical requirements of 3.13.1 through 3.13.5.2 during and after subjection to vibration test of Method 514.1, Curve M, Figure 514.1-3 of MIL-STD-810.

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3.13.1 Sampling plan for examination. - Unless otherwise specified (see

3.12.4 Shock (See 4.7.4). - The network shall not suffer damage, and shall meet the electrical requirements of 3.13.1 through 3.13.5.2 after subjection to shock test of Procedure I, Method 516.1 of MIL-STD-810.

3.12.5 Crash-safety (See 4.7.5). - The network shall withstand an impact of 30G when subjected to crash-safety test of Procedure III, Method 516.1 of MIL-STD-810.

3.12.6 Salt-fog (See 4.7.6). - The network shall be exposed to salt-fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.12.7 Fungus (See 4.7.7). - The network shall show no evidence of viable fungus after being subjected to 28 days of fungus test.

3.13 Electrical (See 4.8). -

3.13.1 Isolation (See 4.8.1). -

3.13.1.1 Input (See 4.8.1.1). - The isolation between the two input ports shall not be less than 20 dB.

3.13.1.2 Output (See 4.8.1.2). - The isolation between the two output ports shall not be less than 18 dB.

3.13.2 Voltage Standing Wave Ratio (VSWR) (See 4.8.2). - The VSWR of each input shall not exceed 1.5:1 with a nominal input impedance of 100 ohms and a nominal output impedance of 50 ohms; all unused ports shall be terminated in their own characteristic impedance.

3.13.3 Insertion loss (See 4.8.3). - The insertion loss shall be measured between each input and output and the insertion loss shall not be greater than 1.1 dB (not including the 3 dB loss due to power division).

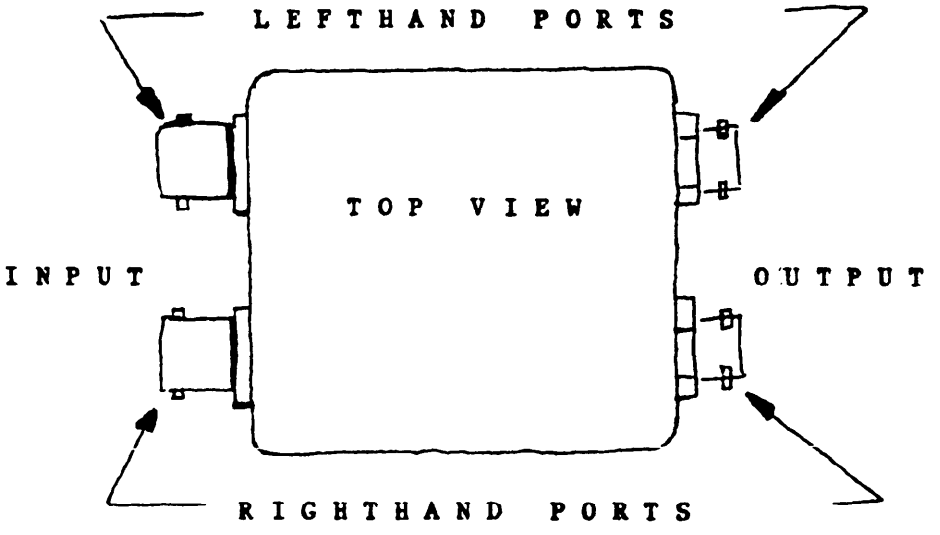
3.13.4 Amplitude balance (See 4.8.4). - The difference between the two output amplitude levels shall not be greater than 0.75 dB.

3.13.5 Phase equality (See 4.8.5). -

3.13.5.1 Input phase (See 4.8.5.1). - The phase difference between the two inputs shall be $90 \pm 7°$.

3.13.5.2 Output phase (See 4.8.5.2). - The phase difference between the two outputs shall be $180 \pm 5°$ for an input to the lefthand port, and $0 \pm 5°$ for an input to the righthand port. (For lefthand and righthand ports identification See Figure 1.)

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CU-1796( )/ARC-114

FIGURE 1

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3.14 Operational inspection (See 4.11). - The network shall be capable of meeting the specified test after applicable processes, inspections and tests have been completed to determine that each unit on order is operable prior to packaging.

3.15 Interchangeability (See 4.10). - Like units shall conform to Requirement 7, of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such item or of the units.

3.16 Technical literature. - Technical literature shall be furnished as specified in the contract.

3.17 Workmanship. - The network shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the following paragraphs herein:

Cleaning	3.5
Securing of parts	3.7
Soldering	3.9
Wiring	3.10
Tropicalization	3.11

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - The inspection requirements specified herein are classsified as follows:

a. First Article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality Conformance inspection.

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(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12)

4.3 First Article. - Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article units. - The contractor shall furnish several First Article units of the complete CU-1796( )/ARC-114.

4.3.2 First Article inspection. - First Article inspection shall consist of the inspection specified in subsidiary documents covering the items listed in 4.4 and the inspections specified for Group A, B and C (See Tables I, II and III respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units; for Group C two units for temperature-altitude, humidity, vibration, shock and crash-safety. One unit each for salt-fog and fungus. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order:

ITEM	WHERE REQUIRED
Finish protective	3.6
Marking	3.8

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performance of any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (See 6.6). Test equipment and facilities for Government verification inspection shall be made available by the contractor.

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4.5.1 Group A Inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table I.

4.5.1.1 Order of inspection within Group A. - Group A inspection shall be performed in an order satisfactory to the Government.

TABLE I - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical CU-1796( )/ARC-114 (Assembled)	3.17	4.9	6.5%	2.5%
Electrical				
Isolation	3.13.1	4.8.1		
Input	3.13.1.1	4.8.1.1		
Output	3.13.1.2	4.8.1.2		
VSWR	3.13.2	4.8.2	*1.0% for	
Insertion loss	3.13.3	4.8.3	combined group.	
Operational inspection	3.14	4.11	0.1%	

*All electrical defects are considered major.

4.5.2 Group B inspection. - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection. This inspection shall conform to Table II and the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. - Group B AQL shall be 6.5 percent defective and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. - Group B inspection shall be performed in an order satisfactory to the Government.

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TABLE II - Group B Inspection

Inspection	Req. Para.	Insp Para.	AQL Major	AQL Minor
Electrical				
Amplitude balance	3.13.4	4.8.4	6.5% for	
Phase equality	3.13.5	4.8.5	combined group.	
Input Phase	3.13.5.1	4.8.5.1		
Output Phase	3.13.5.2	4.8.5.2		
Interchangeability	3.15	4.10	6.5%	

4.5.3 Group C inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspection specified in Table III. Samples shall be selected in accordance with 4.5.4.

TABLE III - Group C Inspection

Inspection	Reqt Para.	Insp Para.
Subgroup I:		
Temperature-Altitude	3.12.1	4.7.1
Humidity	3.12.2	4.7.2
Vibration	3.12.3	4.7.3
Subgroup II:		
Shock	3.12.4	4.7.4
Crash safety	3.12.5	4.7.5
Salt fog	3.12.6	4.7.6
Subgroup III:		
Fungus	3.12.7	4.7.7

4.5.4 Sampling for inspection of equipment. - Units shall be selected at random for each group of inspections in Table III as specified in 4.5.4.1 through 4.5.4.3. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot(s). Selection of samples shall be as specified, and within 45 calendar days after selection, the samples shall have been subjected to and completed the Group C inspections (See 6.4).

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4.5.4.1 Subgroup I. - For this subgroup, two units from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent Group C inspections two units each month shall be selected.

4.5.4.2 Subgroup II. - For this subgroup two units from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units every two months shall be selected.

4.5.4.3 Subgroup III. - For this subgroup, only one unit from the first ten (10) production units shall be selected for the first Group C inspections. For subsequent inspections, one unit shall be selected when one half of the units on order have been accepted by the Government Quality Assurance Representative.

4.5.5 Non-compliance. - For each Group C failure occurrence, the contractor shall make an immediate verbal report to the Government Quality Assurance Representative (QAR) and the cognizant technical activity designated in the contract. Within 3 working days, the contractor shall prepare and forward a written report through the QAR to the cognizant technical activity, with copies to the Administrative Contracting Officer (ACO) and the Procurement Contracting Officer (PCO). The contractor's report shall include but not be limited to the results of his investigation of the cause of failure. Further reports shall cover the results of his further and continuing investigation and details of the proposed corrective action on the processes and materials as applicable, and on all units of product which are manufactured under the same conditions and which the Government considers subjected to the same failure. Within 14 calendar days of his original notice of failure, the contractor shall forward to the cognizant technical activity a final report. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the PCO). Final acceptance and delivery will be withheld until Group C reinspection results have shown that the corrective action was effective and authorized by the PCO.

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Standard test condition. - Unless otherwise specified, all tests shall be performed at ambient temperature, pressure and humidity under the following conditions:

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a. Input impedance 100 ohms.

b. Output impedance 50 ohms.

c. Input and output ports identification, see Figure 1.

4.7 Service conditions. - The service condition tests shall be conducted as specified herein. Upon completion of each service condition test, the equipment shall be subjected to and pass a visual inspection. No defects will be allowed.

4.7.1 Temperature-Altitude. - The network shall be subjected to the temperature-altitude test as specified in Procedure I, Method 504, Table 504-1 of MIL-STD-810. Electrical test shall be performed during and after temperature-altitude to determine compliance with 3.12.1.

4.7.2 Humidity. - The network shall be subjected to the humidity test as specified in Procedure II, Method 507 of MIL-STD-810. Electrical test shall be performed during and after humidity to determine compliance with 3.12.2.

4.7.3 Vibration. - The network shall be subjected to the vibration test as specified in Method 514.1, Curve M, Figure 514.1-3 of MIL-STD-810. Electrical test shall be performed during and after vibration to determine compliance with 3.12.3.

4.7.4 Shock. - The network shall be subjected to the shock test as specified in Procedure I, Method 516.1 of MIL-STD-810. Electrical test shall be performed before and after shock to determine compliance with 3.12.4.

4.7.5 Crash safety. - The network shall be subjected to the crash safety test of Procedure III, Method 516.1 of MIL-STD-810. The requirement of 3.12.5 shall be met. Operational test is not required.

4.7.6 Salt fog. - The network shall be exposed to salt fog test as specified in Method 509 of MIL-STD-810 for 48 hours. The requirement of 3.12.6 shall be met. Operational test is not required.

4.7.7 Fungus. - The network shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. The requirement of 3.12.7 shall be met. Operational test is not required.

4.8 Electrical. -

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4.8.1 Isolation. -

4.8.1.1 Input. - Apply a signal to the output and sweep the frequency band from 30 to 76 MHz and measure between the two input ports to determine compliance with 3.13.1.1.

4.8.1.2 Output. - Apply a signal to the input and sweep the frequency band from 30 to 76 MHz and measure between the two output ports to determine compliance with 3.13.1.2.

4.8.2 Voltage Standing Wave Ratio (VSWR). - With an input impedance of 100 ohms and an output impedance of 50 ohms, terminate the unused ports in their own characteristic impedance and sweep the frequency band from 30 to 76 MHz to determine compliance with 3.13.2.

4.8.3 Insertion loss. - Apply a signal to the input port and sweep the frequency band from 30 to 76 MHz and measure the insertion loss at the corresponding output ports to determine compliance with 3.13.3.

4.8.4 Amplitude balance. - Apply a signal of 76 MHz to the input and measure the amplitude levels between the two output ports to determine compliance with 3.13.4.

4.8.5 Phase equality. -

4.8.5.1 Input phase. - Apply a signal of 76 MHz to either output port and measure the phase difference between the input ports to determine compliance with 3.13.5.1.

4.8.5.2 Output phase. - Apply a signal of 76 MHz to the left input port and measure the output. Apply a signal of 76 MHz to the right input port and measure the output. The requirement of 3.13.5.2 shall be met.

4.9 Visual-Mechanical. - Prior to and after assembly, the network shall be examined for defects in accordance with MIL-STD-252.

4.10 Interchangeability. - The dimensions listed below shall be measured to determine compliance with the physical interchangeability requirements of 3.15. When a listed dimension is not within specified or design limits, it shall be considered a major defect:

a. External and internal dimensions of moldings, mounting cover, cases and insertable parts, when such dimensions affect mating.

b. Dimension of cavities, when such dimensions affect insertable items.

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c. Location of holes on separable parts which must mate such as mounting assemblies.

d. Location of connectors, mounting bracket holes.

e. Size and form of thread.

4.11 Operational inspection. - The network shall be inspected and tested as specified in 4.8.2 to determine compliance with 3.14.

4.12 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the packaging requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C as specified (See 6.2).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Network Impedance Matching, Quadrature Hybrid, CU-1796( )/ARC-114 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Network Impedance, Matching Quadrature Hybrid, CU-1796( )/ARC-114 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Each Network Impedance, Matching Quadrature Hybrid, CU-1796( )/ARC-114 shall be individually packaged Method III in accordance with MIL-P-116 as follows: Cushion each item by wrapping in paperboard conforming to PPP-P-291, Type 1 and secure with tape conforming to PPP-T-45, Type III, Grade A. Place the cushioned item within a close-fitting box conforming to MIL-B-43014, Form I, Style II. Closure shall be in accordance with the appendix of the box specification.

5.1.1.5 Intermediate package. - A quantity of 50 each Network Impedance, Matching Quadrature Hybrid, CU-1796( )/ARC-114 packaged as specified in 5.1.1.4, shall be placed within a close fitting box conforming to MIL-B-43014, Form I, Style II. Closure shall be in accordance with the Appendix of the box specification.

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5.1.2 Level C. - Network Impedance, Matching Quadrature Hybrid, CU-1796( )/ARC-114, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be Level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2)

5.2.1 Level A. -

5.2.1.1 Consolidation. - A quantity of Network Impedance, Matching quadrature Hybrid, CU-1796( )/ARC-114, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, Type CF, class weather-resistant. Box closure shall be as specified in the Appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load Type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, Type IV, Size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per designation are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the Appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, Style 4, Class 2; or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 58 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be appplied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or Appendix thereto except that metal strapping shall conform to QQ-S-781, Type 1, Class B.

5.2.2 Level B. -

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5.2.2.1 Consolidation. - A quantity of Network Impedance, Matching Quadrature Hybrid, CU-1796( )/ARC-114, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, Type IV as specified in the Appendix of the box specification. No further packing shall be required.

5.2.3 Level C. -

5.2.3.1 Consolidation. - A quantity of Network Impedance, Matching Quadrature Hybrid, CU-1796( )/ARC-114, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The CU-1796( )/ARC-114 is intended to be used as a broadband impedance matching device and to compare radio frequency signals of the same frequency applied to its inputs; and provides signals out of its outputs 90° out of phase to Radio Set AN/ARC-114 homing inputs when installed in Helicopter OH-6.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

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b. Level of preservations, packaging and level of packing required for shipment.

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. First Article inspection. - Several samples of each item cited in Section 1 are required so that lengthy environmental tests can be completed on sample units while complete performance measurements can be made on other unit(s). (See 3.4)

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required. (See 3.16)

h. Quality of tools and running spare parts required.

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CU-1796W/ARC-114. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending results of the corrective action authorized by the cognizant technical activity designated in the contract.

6.5 Definition of Level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less and distribution within the Continental Limits of the United States.

6.6 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive Quality Assurance Provisions established in the specification and contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian
Army EL

Preparing Activity
Army EL
Project No. 5915-A026

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-N-55374(EL) NETWORK, IMPEDANCE MATCHING QUADRATURE HYBRID CU-1796( )/ARC-114

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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